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# Particle in cell simulations of tearing modes in reversed-field-pinch-like plasma

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## Abstract

Particle in cell (PIC) simulations of tearing modes in two-dimensional plane geometry in a force free reversed field pinch (RFP) like plasma equilibrium are performed to study possible kinetic effects on these modes in RFPs. Linear tearing modes are compared in the PIC and two fluid models. The results showed that the growth rates and the profiles of magnetic field components in the two models are very similar, indicating that the kinetic effects on the tearing modes are weak such that the two fluid approximation is rather accurate for modeling these instabilities in RFPs. During the nonlinear evolution of the tearing mode in this geometry small scale secondary instabilities located near the internal layer of the primary tearing instability are excited. These secondary instabilities appear to be driven by the nonlinearly induced local pressure gradient in the regions of unfavorable curvature of the nonlinearly evolved magnetic field. They could also appear in a realistic RFP geometry and play a role during sawtooth crashes in these machines.

## I. INTRODUCTION

Tearing instabilities play an important role in laboratory, space and astrophysical plasmas, see e. g. Ref. [1]. They are thought to be responsible for fast reconnection of magnetic fields, relaxation to the Taylor state and the dynamo effect in reversed field pinch (RFP)[2]-[6]. It is also believed that the tearing instabilities contribute to the ion heating [7] and the momentum transport [8] in RFPs, observed during internal reconnection events (sawtooth crashes) [9]-[16] in these machines. The understanding of the physics of tearing instabilities is important for the understanding of various processes in RFP.

The most common models used to study tearing instabilities are single fluid resistive magnetohydrodynamics (MHD) [17],[18] and two fluid MHD [19]-[23]. Recent two fluid modeling of tearing modes in RFP plasma [21],[24],[25] showed that the two fluid effects significantly change the modes parameters in comparison with the parameters calculated with the resistive MHD model. The two fluid effects modify both the modes growth rate and its polarization. Analysis of the size of internal layer in a tearing mode relevant to RFP plasma [7] showed that the spatial scale of velocity, electric field and current profiles near a resonance surface is comparable to the ion gyroradius. The latter is relatively large in RFPs because of the smaller equilibrium magnetic field. Thus, in addition to the two fluid effects, the effects due to the variation of field components of the tearing mode within the ion gyroradius, finite Larmor radius (FLR) effects, or other kinetic effects, can influence these modes in RFP as well.

In this study we analyze possible influence of the kinetic effects on linear tearing modes in RFP-like plasma by comparing these modes calculated in a fully kinetic collisionless particle-in-cell (PIC) model and in the two fluid model. First principles time evolving particle-in-cell code VPIC [26] is used in the kinetic modeling. Because of practical limitations of the PIC simulations we consider the tearing modes in plane geometry in a force free RFP-like plasma equilibrium bounded by two conducting walls. In this model plasma perturbations depend on $x$ and $y$ coordinates while $z$ coordinate is ignorable, such that a two-dimensional (in configuration space) version of VPIC is used. Our results showed that the growth rates and profiles of magnetic field components of the tearing modes in the two fluid and in the kinetic models are very similar except for some additional broadening of the PIC profiles near the resonance surface, indicating that the kinetic effects on the tearing modes in RFP are weak.

While the amplitudes of isotropic and anisotropic parts of the pressure tensor calculated from PIC results have similar magnitudes, the contribution to the fluid force from the isotropic part of the pressure tensor significantly exceeds the one from the anisotropic part making the fluid approximation to be rather accurate for modeling of the tearing modes in RFP.

In addition to the analysis of the linear tearing modes in the fully kinetic model, we use VPIC to study nonlinear time evolution of the tearing modes in this geometry. The use of first principles PIC code is a powerful tool in the search for new effects which can be relevant to RFP discharges and which could be undetected by the present experimental diagnostics. In contrast to a cylindrical model, which requires 3-D simulations, there is no self-maintained (by dynamo) RFP-like equilibrium in the plane geometry. In our simulations an initially imposed unstable RFP-like equilibrium is transformed during the nonlinear evolution of the tearing mode into a new equilibrium which is topologically different from the initial equilibrium. Due to this reason we concentrate on the initial stage of the nonlinear mode evolution to find possible new effects that can be relevant to the tearing modes in RFP. In our simulations we observed the excitation of small scale secondary instabilities located near the internal layer of the primary tearing instability. We identify these secondary instabilities as the ones driven by the nonlinearly induced local pressure gradient in the regions of unfavorable curvature of the nonlinearly evolved magnetic field. Such secondary instabilities can also appear in a realistic RFP geometry and play a role during sawtooth crashes in these machines. More detailed studies of these instabilities are the subject of the future work.

The paper is organized as follows. In Sec. II plasma geometry, its equilibrium and simulation parameters are discussed. In Sec. III the results of PIC simulations are presented and analyzed. We summarize in Sec. IV.

## II. PLASMA MODEL

We consider plasma in plane geometry occupying region $0 \leq x \leq 2a$. Conducting walls are located at $x = 0$ and $x = 2a$. Plasma is uniform in $y$ and $z$ coordinates. Throughout the paper we use normalization

$$x = a\tilde{x}\,,\ \ v = v_A\tilde{v}\,,\ \ t = \tau_A\tilde{t}\,,\ \ B = B_0\tilde{B}\,,\ \ j = \frac{c}{4\pi}\frac{B_0}{a}\tilde{j}\,,\ \ n = n_0\tilde{n}\,,\ \ \beta = \frac{8\pi p_0}{B_0^2}\,,$$

where $v_A$ is the Alfvén speed, $\tau_A = a/v_A$, $B_0$ is the equilibrium magnetic field, $n_0$ is the equilibrium density of electrons or ions and $p_0$ is the equilibrium plasma pressure.

In this geometry there are RFP-like force free equilibria with uniform pressure and density profiles and zero plasma velocity. Such equilibrium magnetic field satisfies

$$\nabla \times \bar{\boldsymbol{B}} = \lambda \bar{\boldsymbol{B}} . \tag{1}$$

The function $\lambda(x)$ specifies the equilibrium current profile. We specify this profile in the form similar to the one used in cylindrical RFP geometry [2],

$$\lambda(x) = 2\Theta \left[1 - (x-1)^{\alpha}\right] . \tag{2}$$

Equations (1) and (2) lead to equations for the components $\bar{B}_y$ and $\bar{B}_z$ of the equilibrium magnetic field which are subject to the boundary conditions $\bar{B}_y(1) = 0$ and $\bar{B}_z(1) = 1$. Solution of these equations define equilibrium magnetic field in the specified force free equilibrium. In the considered plane geometry the magnitude of this field is independent of $x$, $\bar{B}_y^2 + \bar{B}_z^2 = 1$. In our simulations we use $\Theta = 2.6$, $\alpha = 1.3$ which results in the equilibrium field components shown in Fig. 1. As in the case of cylindrical RFP equilibrium, the component $\bar{B}_z$ reverses at some distance along the $x$ axis. In our two-dimensional model we consider solutions of plasma equations which are independent of the $z$ coordinate and apply periodic boundary conditions in the $y$ direction. In this case the component of wave number of plasma perturbations parallel to the equilibrium magnetic field is $k_{||} = k_y \bar{B}_y$. Thus there is only one resonant surface, where $k_{||} = 0$, for perturbations around this equilibrium located at $x = 1$. Tearing modes with different wave numbers $k_y$, resonant at this location, are unstable in this equilibrium. This configuration is used to approximate tearing modes in a more realistic RFP geometry.

In the PIC simulations the initial electron distribution function is taken to be a shifted Maxwellian with the drift velocity $\boldsymbol{v}_e$ corresponding to the equilibrium plasma current, $\boldsymbol{j} = -n_e e \boldsymbol{v}_e$. The initial ion distribution function is Maxwellian with the same temperature as the electron component. The specified initial magnetic field and particles distribution functions are not in exact kinetic equilibrium because of the finite shear of the equilibrium magnetic field. But since the shear is relatively small on a scale of electron Larmor radius, the use of these initial conditions is justified in these simulations. In spite of using simplified geometry in our studies, fully kinetic PIC simulations still present a computational challenge for RFP plasma parameters. In order to make the simulations practical we use somewhat different plasma parameters in comparison with the RFP plasma. In our simulations $a =$

10 cm, $B_0 = 30\,\mathrm{kG}$, $n_e = 10^{13}\,\mathrm{cm}^{-3}$, $\beta = 0.25$ and $m_i/m_e = 100$. This corresponds to $d_e = c/\omega_{pe} = 0.72\,\mathrm{cm}$, $d_i = c/\omega_{pi} = 7.2\,\mathrm{cm}$, $\rho_{Li} = 1.8\,\mathrm{cm}$. For the selected parameters the equilibrium magnetic field is an order of magnitude larger than in RFP. This restriction is imposed by the requirement to resolve the Debye length scale in simulations (the latter increases with the increase of $B_0$ for fixed $\beta$). Because of the relatively high plasma beta the ion Larmor radius is comparable with the width of the tearing layer which allows to study possible kinetic effects on tearing modes in this configuration.

In this geometry field components of the linear tearing mode are proportional to $\exp(\gamma t + ik_y y)$. For these plasma parameters the dispersion curves of growing modes in the two fluid model with zero resistivity are shown in Fig. 2. This result is calculated by solving eigenmode problem using algorithm described in Ref. [25]. The result is symmetric when $k_y$ is changed to $-k_y$; thus the dispersion curves are shown only for positive values of $k_y$. The solid line corresponds to the tearing mode branch. The dashed line branch is associated with the linearized term $(\boldsymbol{v}_e\boldsymbol{\nabla})\boldsymbol{v}_e$ in the two fluid equations which contributes due to the nonzero equilibrium velocity of the electron component. This branch disappears when this term is not included in the equations. For the selected equilibrium current profile the instability drive of the tearing mode is strong, such that the maximum growth rate, reached for $k_y a \approx 2$, is comparable with the inverse Alfvén time. This highly unstable equilibrium is used to reduce the total simulation time to make the PIC simulations more practical.

In the PIC simulations the $y$ coordinate belongs to the range $0 \leq y \leq \pi a$ and the periodic boundary conditions are used along this axis, such that the $y$ coordinate approximates poloidal direction in RFP. Because of the periodic boundary conditions the $y$ dependence of field components can be expanded in Fourier harmonics proportional to $\exp(i2my)$, with $m = 0, \pm 1, \pm 2, \ldots$. If the linear tearing modes in the PIC model are not very different from the ones in the two fluid model then the dominant contribution to the PIC eigenmode is expected to be from the $|m| = 1$ harmonics, which corresponds to $k_y a = 2$ in Fig. 2. The modes corresponding to other harmonics in PIC simulation have much smaller growth rates. Thus, selection of this size of the domain in the $y$ direction results in only one dominant mode in the simulations which simplifies the mode analysis.

## III. RESULTS OF PARTICLE IN CELL SIMULATIONS

PIC simulations are carried out on a multi processor supercomputer using approximately $10^9$ particles. The tearing eigenmode growth from the computational noise present in the PIC model. The time evolution of volume integrated contributions to magnetic energy and kinetic energy of electrons and ions is presented in Fig. 3. Each quantity is normalized on the initial volume integrated magnetic energy. The total energy (electromagnetic plus kinetic) in the system is conserved with a very good accuracy. The initial kinetic energy of electrons contains the contribution due to their drift motion in the equilibrium current. Some initial oscillations in these time dependencies are because the system is not in an exact kinetic equilibrium at $t = 0$. Equilibrium magnetic field has only $\bar{B}_y$ and $\bar{B}_z$ components such that the initial magnetic energy in the $B_x$ component is zero in this figure. The dominant tearing mode corresponds to the $m = 1$ harmonics. This mode grows first linearly and then nonlinearly until the system saturates in a new quasi equilibrium. The transition from the initial to the final state is accompanied by a strong modification of the magnetic field geometry along with significant compression and rarefication of plasma in different parts of the domain. In the final state plasma is almost stationary (no ion flows) and the increase of kinetic energies of ions and electrons during the transition is due to compressional plasma heating in the regions with the increased pressure.

During the linear evolution the mode grows exponentially with the growth rate $\gamma\tau_A = 0.75$ which is close to the growth rate of this mode calculated in the two fluid model, $\gamma\tau_A = 0.7$ for $k_y a = 2$ in Fig. 2. Comparison of magnetic field components in the linear tearing eigenmode in the two models is made in Fig. 4. The data for the PIC results are taken at $t/\tau_A = 2.8$ when the mode experiences linear growth. Field components are in dimensionless units, they are normalized such that $B_x(1) = 1$. This comparison shows that the field profiles are similar in the two models except for the profiles calculated in the PIC model being somewhat wider near the resonance surface $x = 1$. This noticeable broadening of the PIC eigenfunctions can be related to the FLR effects, the ion Larmor radius is a sizable portion of the tearing layer width in these simulations.

In the considered collisionless plasma model the kinetic effects enter the fluid momentum equation of plasma species $\alpha$ in terms of the gradient of the pressure tensor $\hat{\boldsymbol{P}}_\alpha$, see e. g.

Ref. [27]. We decompose this tensor into scalar and tensor parts as

$$\hat{\boldsymbol{P}}_\alpha = p_\alpha \hat{\boldsymbol{I}} + \hat{\boldsymbol{\pi}}_\alpha\,, \quad \text{where } p_\alpha = \frac{1}{3}\left(P_{\alpha xx} + P_{\alpha yy} + P_{\alpha zz}\right)$$

and $\hat{\boldsymbol{I}}$ is a unit tensor. Assumption that the pressure is scalar is a typical approximation in two fluid plasma models. It is used in our two fluid model here. The magnitudes of the linearized fluid forces due to the scalar and tensor parts of the pressure tensor for the considered tearing mode in the PIC model are shown in Fig. 5. All profiles in this figure are normalized on the same value. This comparison shows that the force due to the scalar pressure dominates in the total force due to gradient of the pressure tensor which explains the closeness of the tearing mode parameters calculated in the two fluid and PIC models. One should note that the scalar and tensor parts of the pressure tensor are comparable in magnitude, but their gradients, contributing to the fluid force, differ significantly. The results of our simulations show that the two fluid model is a good approximation in modeling of tearing modes in RFPs.

We analyze nonlinear time evolution of the tearing instability in this geometry. Profiles of different plasma quantities in $xy$ plane at the moment of time $t/\tau_A = 3.81$ are shown in Figs. 6(a)-(f). Their normalization is defined in the beginning of Sec. II. Nonlinear effects have influenced the mode structure by this moment of time. The $m = \pm 1$ harmonics are the dominant plasma perturbations in this figure. Magnetic field component $B_x$ reached approximately 6% of the magnitude of the equilibrium magnetic field, Fig. 6(a). Current sheet like perturbation, typical of a tearing mode, can be seen in the $j_z$ profile near the resonance surface $x = 1$ in Fig. 6(d). The amplitudes of ion velocity and magnetic field perturbations in these dimensionless units are comparable to each other, see Figs. 6(a) and 6(e). This result means that the plasma flow velocity in the tearing mode at a sawtooth crash in RFP can be a sizable portion of the ion thermal velocity which can contribute to the ion heating. Our result confirms similar result obtained in resistive MHD plasma model in Ref. [7] in which the ion heating by viscous damping of plasma flows was examined. Figure 6(f) shows that the plasma density perturbation reached $\sim 15\%$ of its equilibrium value at this moment of time. Plasma quasineutrality is approximately conserved in these simulations.

In addition to the dominant $m = \pm 1$ perturbations related to the considered tearing mode, a spectrum of perturbations with large wave numbers peaked near $|m| \sim 12$ is also

present in this figure. These small scale perturbations are clearly seen in Figs. 6(e) and 6(f). These are the secondary instabilities, they are not excited when the primary tearing mode is stable. This was checked in simulations with the reduced size of the domain in $y$ direction in which all $|m| \geq 1$ tearing modes are stable ($k_y a$ for these modes are larger than the tearing mode marginal point in Fig. 2). The growth rates of these secondary instabilities are comparable to the growth rate of the primary tearing mode in our simulations. We identify these secondary instabilities as driven by the pressure gradient in the regions of unfavorable curvature of the magnetic field [28]. The pressure gradient and magnetic field curvature are created during the nonlinear evolution of the primary tearing mode. Figure 7 shows projections of magnetic field lines on the $xy$ plane plotted on top of the plasma density profile at the same moment of time as in Fig. 6. This figure shows that the small scale instabilities are excited in places where there is a particular relative sign of the magnetic field line curvature with respect to the local density (pressure) gradient corresponding to the usual unfavorable curvature convention [28].

We also modeled these secondary instabilities in cylindrical RFP equilibrium in ideal MHD model by imposing a small pressure peak near a resonance surface on top of the uniform equilibrium pressure profile with the peak's parameters similar to the local maxima of plasma density in Fig. 7. In this model the equilibrium magnetic field supplies the unfavorable curvature. In this analysis a spectrum of unstable modes with the growth rates and perpendicular wave numbers in the range similar to the observed in these secondary instabilities is found. This simplified modeling supports our conclusion about the source of the observed secondary instabilities in the PIC simulations. These instabilities can appear in realistic RFP geometry when the local pressure gradient is created by the tearing mode. The unfavorable curvature is already embedded in the magnetic equilibrium in this case. Because these instabilities are accompanied by small scale plasma velocity perturbations, see Fig. 6(e), they can contribute to the ion heating observed at sawtooth crashes in RFPs. More detailed studies of possible effects of these instabilities on RFP plasma is subject of the future work.

## IV. SUMMARY AND CONCLUSIONS

Simulations of tearing modes in two-dimensional plane geometry in RFP-like plasma equilibrium are performed in the PIC model. Parameters of linear tearing eigenmodes in the PIC and two fluid plasma models in these geometry are compared. The results showed that the growth rates and profiles of magnetic field components in the two models are very similar except for some broadening of the PIC tearing mode near the resonance surface when compared with the two fluid mode. Thus the kinetic effects on the tearing modes are relatively weak and the two fluid approximation is rather accurate for modeling these instabilities in RFPs.

Nonlinear time evolution of the tearing mode in the PIC model is also studied in this geometry. The conclusion, made in previous studies based on resistive MHD model, that the plasma flow velocity in nonlinearly evolved tearing mode can be a sizable portion of the ion thermal velocity is confirmed in our results. Small scale secondary instabilities, located near the internal layer of the primary tearing instability, are found in our simulations. These secondary instabilities are driven by the nonlinearly induced local pressure gradient in the regions of unfavorable curvature of the nonlinearly evolved magnetic field. They can also appear in a realistic RFP geometry and play a role during sawtooth crashes in these machines. More detailed studies of these instabilities are subject of the future work.

## ACKNOWLEDGMENTS

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FIG. 1: Components of equilibrium magnetic field. $\Theta = 2.6$, $\alpha = 1.3$.

FIG. 2: $\gamma\tau_A$ vs $k_y a$ for growing modes in two fluid model.

FIG. 3: (Color online) Time evolution of volume integrated contributions to magnetic energy and kinetic energy of electrons and ions.

FIG. 4: Components of magnetic field in tearing mode in two fluid model (a)-(c) and from PIC simulations (d)-(f). $\Theta = 2.6$, $\alpha = 1.3$, $k_y a = 2$, $m_i/m_e = 100$, $\beta = 0.25$, $d_e/a = 0.072$, $d_i/a = 0.72$, $\rho_{Li}/a = 0.18$.

FIG. 5: Contributions to fluid force from scalar and tensor parts of the pressure tensor for (a) electrons and (b) ions.

FIG. 6: (Color) Profiles of (a) $B_x$, (b) $B_y$, (c) $B_z$, (d) $J_z$, (e) $V_{iy}$, (f) $\rho_i$ in $xy$ plane at $t/\tau_A = 3.81$.

FIG. 7: (Color) Ion density distribution and projections of magnetic field lines on $xy$ plane at $t/\tau_A = 3.81$.

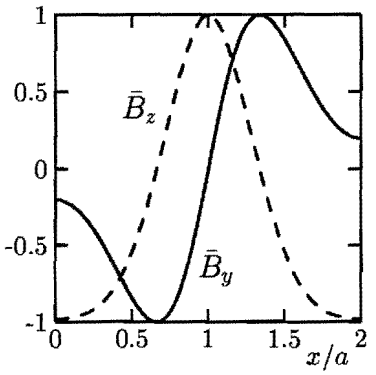



Figure 1: Components of equilibrium magnetic field. $\Theta = 2.6$, $\alpha = 1.3$.

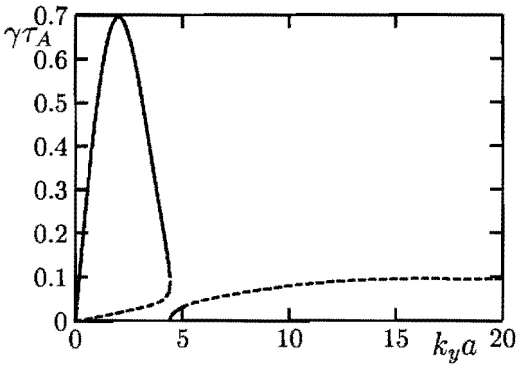



Figure 2: $\gamma\tau_A$ vs $k_y a$ for growing modes in two fluid model.

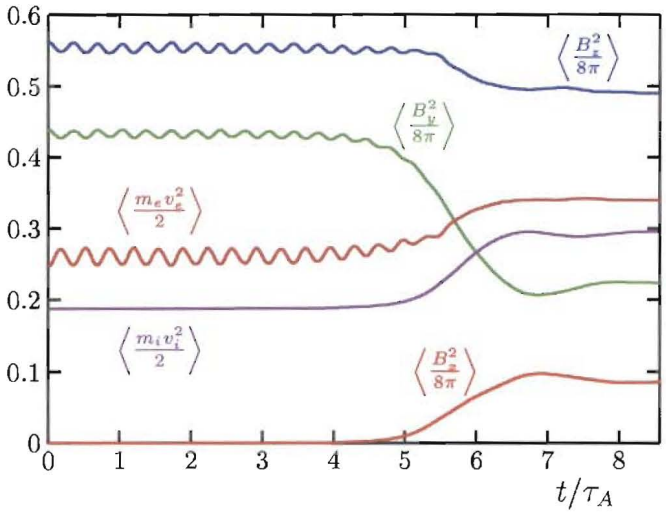



Figure 3: (Color online) Time evolution of volume integrated contributions to magnetic energy and kinetic energy of electrons and ions.

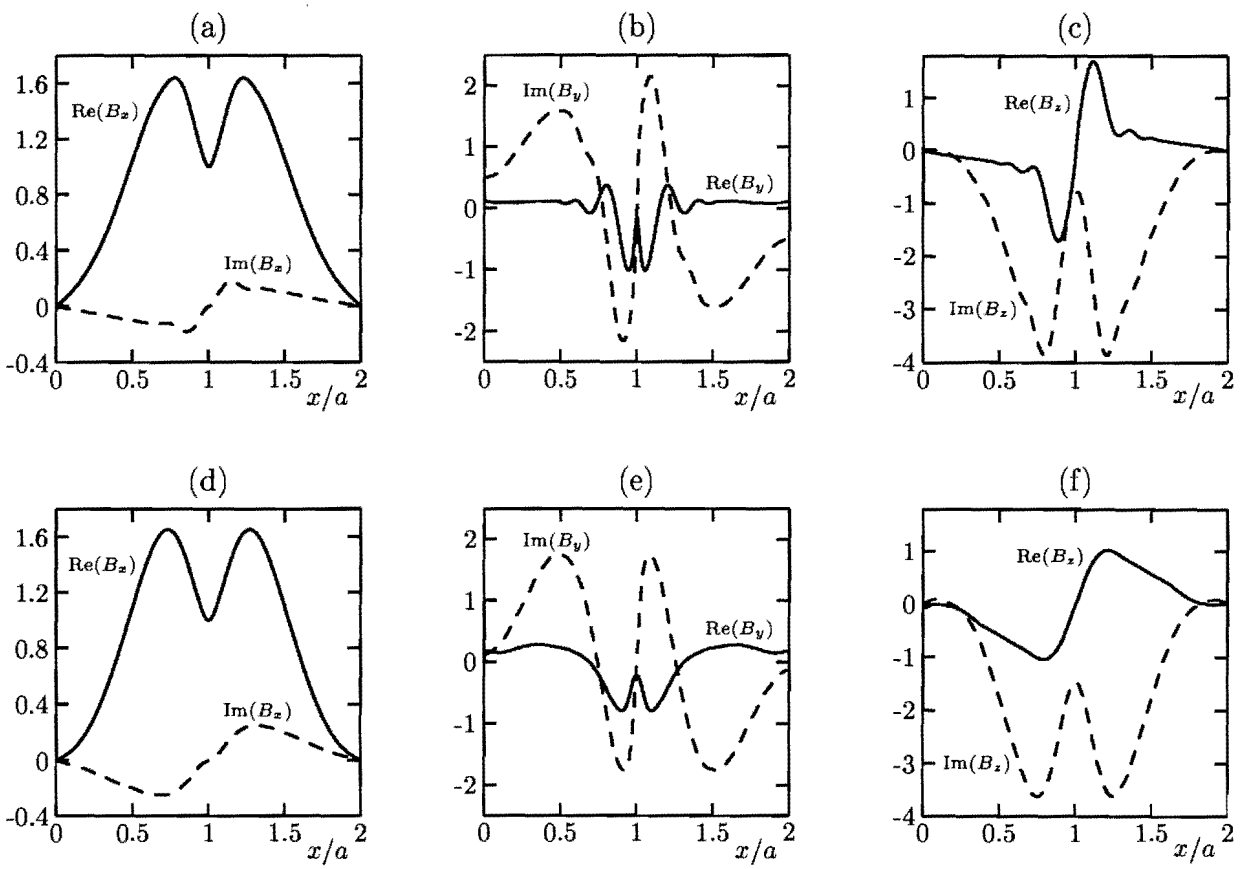



Figure 4: Components of magnetic field in tearing mode in two fluid model (a)-(c) and from PIC simulations (d)-(f). $\Theta = 2.6$, $\alpha = 1.3$, $k_y a = 2$, $m_i/m_e = 100$, $\beta = 0.25$, $d_e/a = 0.072$, $d_i/a = 0.72$, $\rho_{Li}/a = 0.18$.

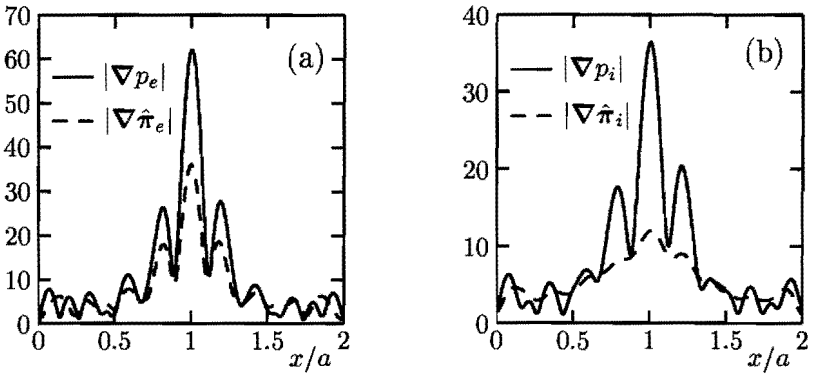



Figure 5: Contributions to fluid force from scalar and tensor parts of the pressure tensor for (a) electrons and (b) ions.

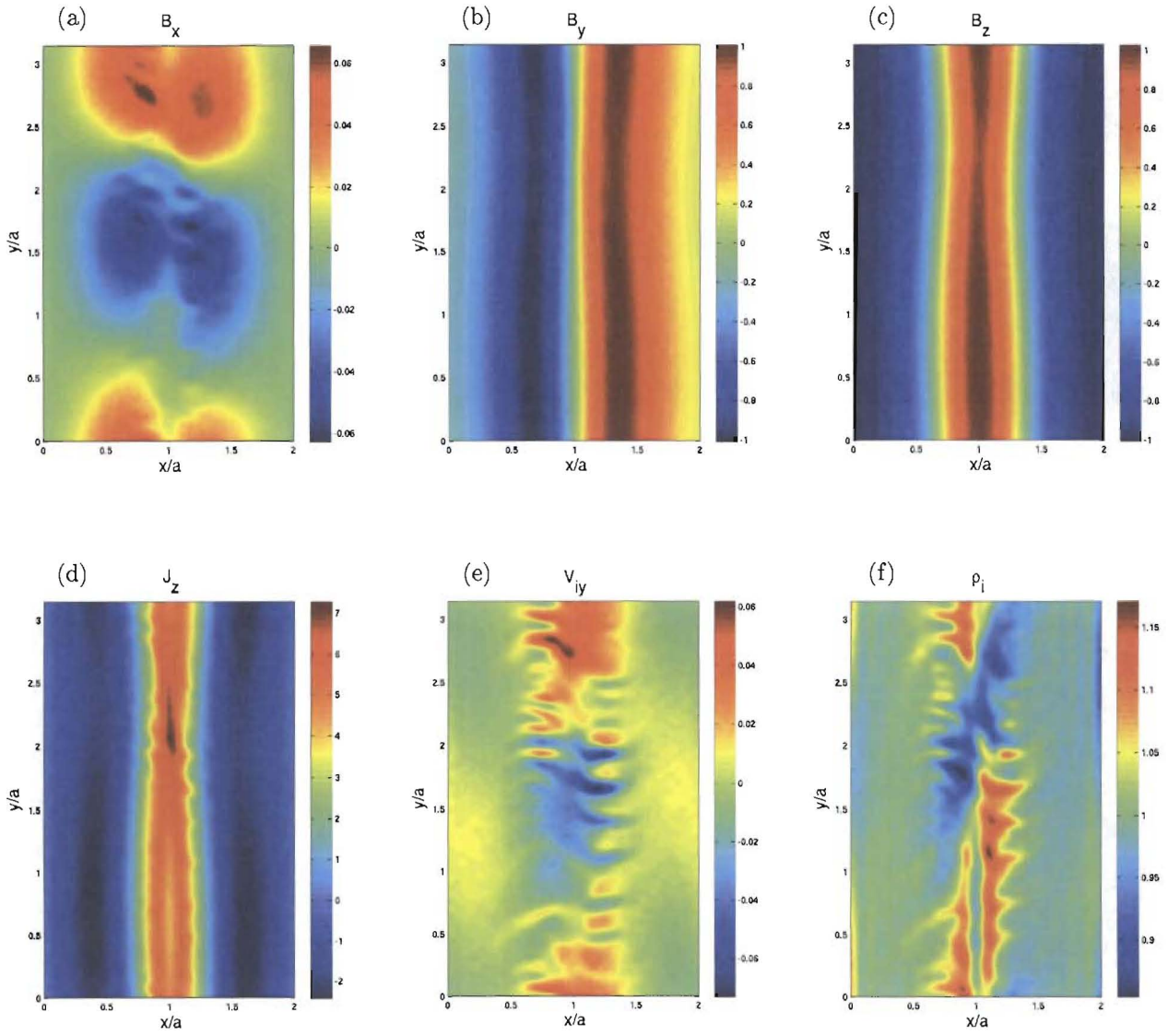



Figure 6: (Color) Profiles of (a) $B_x$, (b) $B_y$, (c) $B_z$, (d) $J_z$, (e) $V_{iy}$, (f) $\rho_i$ in $xy$ plane at $t/\tau_A = 3.81$.

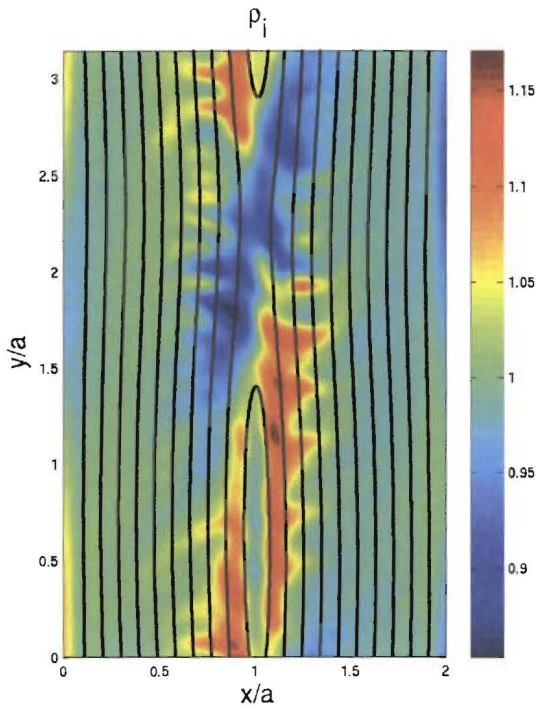



Figure 7: (Color) Ion density distribution and projections of magnetic field lines on $xy$ plane at $t/\tau_A = 3.81$.